

CEP Mapping Tables For Multifunctional Fireguard Bot:

CEP Mapping: Ks are addressed through the Project

Ks Attribute	How Addressed in This Project	CLOs	PLOs
K3 - Engineering Fundamentals	GPIO pin configuration on ESP32-CAM and Arduino Uno; PWM signal generation for L298N motor driver; SoftwareSerial communication at 9600 baud between two MCUs	CLO1, CLO2, CLO5	a, c
K4 - Specialist Knowledge	ESP32-CAM firmware for SoftAP Wi-Fi hosting, MJPEG camera streaming, and HTTP web server; Arduino firmware for real-time IR (D4, D5) and flame sensor (D6, D7, D8) polling; buzzer alarm (D3)	CLO1, CLO2, CLO5	a, c
K5 - Knowledge for Design	Dual Primary-Secondary controller architecture design; four-mode state machine (Manual, Line Follower, Fire Detection, YOLO AI); power budgeting with 18650 Li-ion batteries; responsive web dashboard at http://192.168.4.1	CLO3, CLO5	b, c
K6 - Engineering Practice	Physical hardware assembly and wiring; calibration of IR and flame sensors on prototype chassis; integration testing across all four operational modes; emergency stop validation	CLO1, CLO2, CLO5	a, c

CEP Mapping: Ps are addressed through the Project

Ps Attribute	How Addressed in This Project	CLOs	PLOs
P1 - Depth of Knowledge	Simultaneous real-time sensor fusion (IR + triple flame sensors) on Arduino while ESP32-CAM handles Wi-Fi, camera stream, and motor PWM — requires fundamentals-based interrupt and timing design	CLO1	a
P2 - Conflicting Requirements	Real-time sensor polling speed vs. Wi-Fi web server responsiveness; fire detection sensitivity vs. false alarm avoidance; PWM speed control range (50–255) vs. safe motor operation; low cost vs. multi-mode performance	CLO1, CLO2	a, c

P7 - Interdependence	Flame detection (Arduino) → motor stop → buzzer alarm; IR reading → serial command → ESP32 PWM actuation; YOLO inference on laptop → HTTP GET → ESP32 motor drive — all sub-systems tightly coupled via SoftwareSerial and Wi-Fi	CLO1, CLO3	a, b
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CEP Mapping: As are addressed through the Project

As Attribute	How Addressed in This Project	CLOs	PLOs
A1 - Range of Resources	Selection and integration of ESP32-CAM, Arduino Uno R3, L298N motor driver, 4× DC gear motors, 2× IR sensors, 3× flame sensors, active buzzer, SG90 servo (wired, not activated), water pump relay (wired, not activated), and 18650 Li-ion batteries	CLO1, CLO2, CLO5	a, c
A2 - Level of Interaction	Four interacting sub-systems coordinated: sensing unit (Arduino Uno), actuation unit (L298N + 4 motors), wireless communication unit (ESP32-CAM SoftAP), and AI navigation unit (laptop YOLOv11 + OpenCV via HTTP)	CLO1, CLO2, CLO5	a, c
A5: Familiarity	Extending beyond standard lab experiments by integrating a local microcontroller system with a Python-based YOLOv11 AI vision script over a custom wireless network.	CLO1, CLO2, CLO6	a, c, j